



# Introduction to Brain and Cognition

S. Bapi Raju

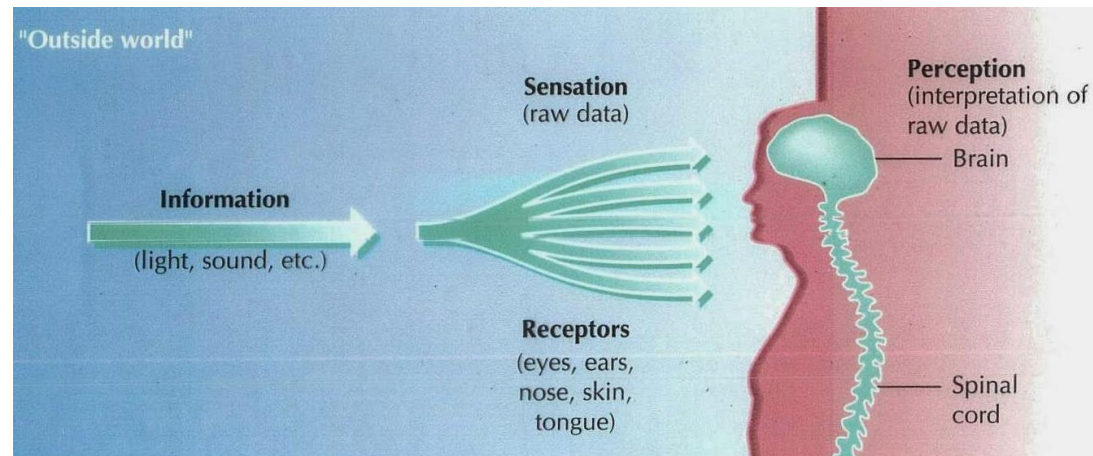
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# Sensation & Perception

# SENSATION



- gathering info from the environment via your senses
- senses pick up visual, auditory, and other sensory stimuli and transmit them to the brain
- sensory information registers in the brain but has not been interpreted



# PERCEPTION

*the* ILLUSION  
*of* REALITY

- process by which sensory information is actively organized and interpreted by the brain
- understanding what is being sensed
- brain's best guess of the cause of sensory inputs

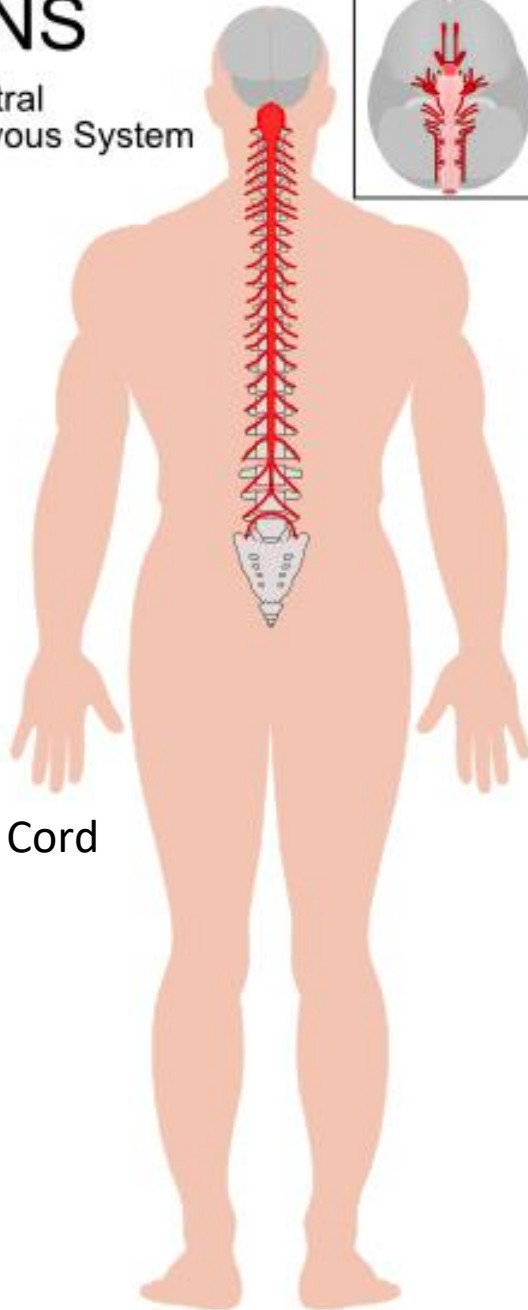


# SENSATION



# CNS

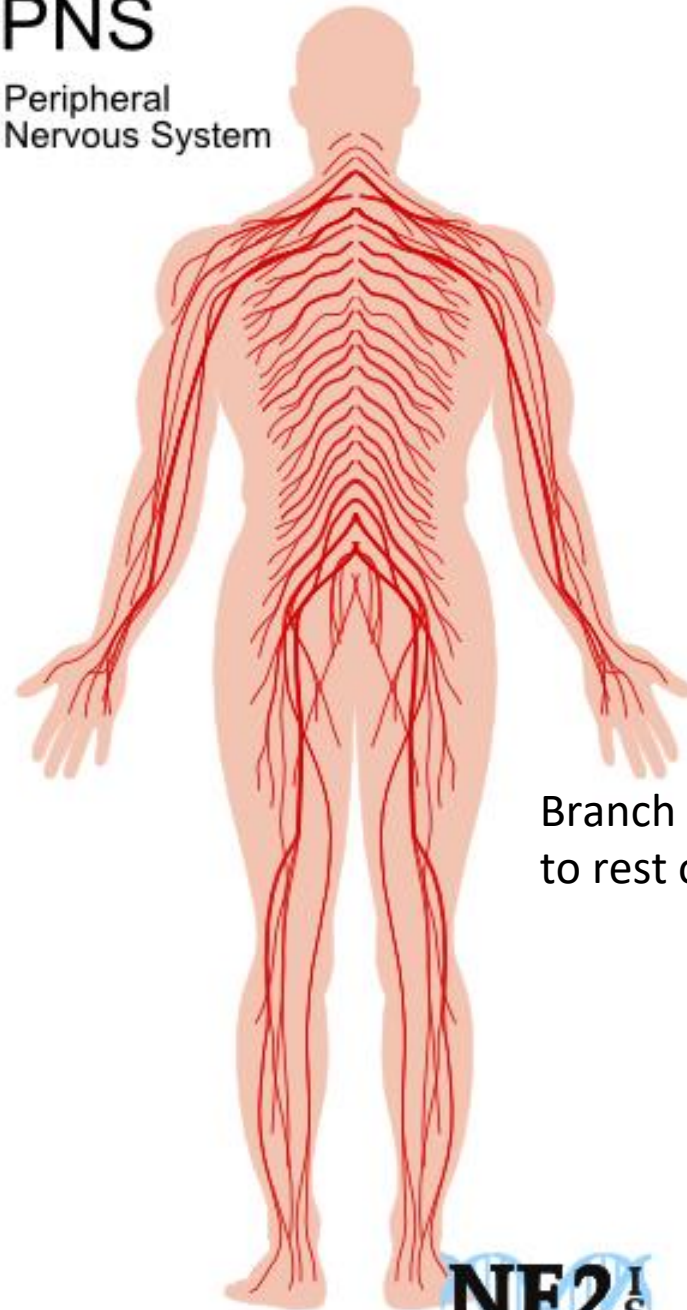
Central  
Nervous System



Brain & Spinal Cord

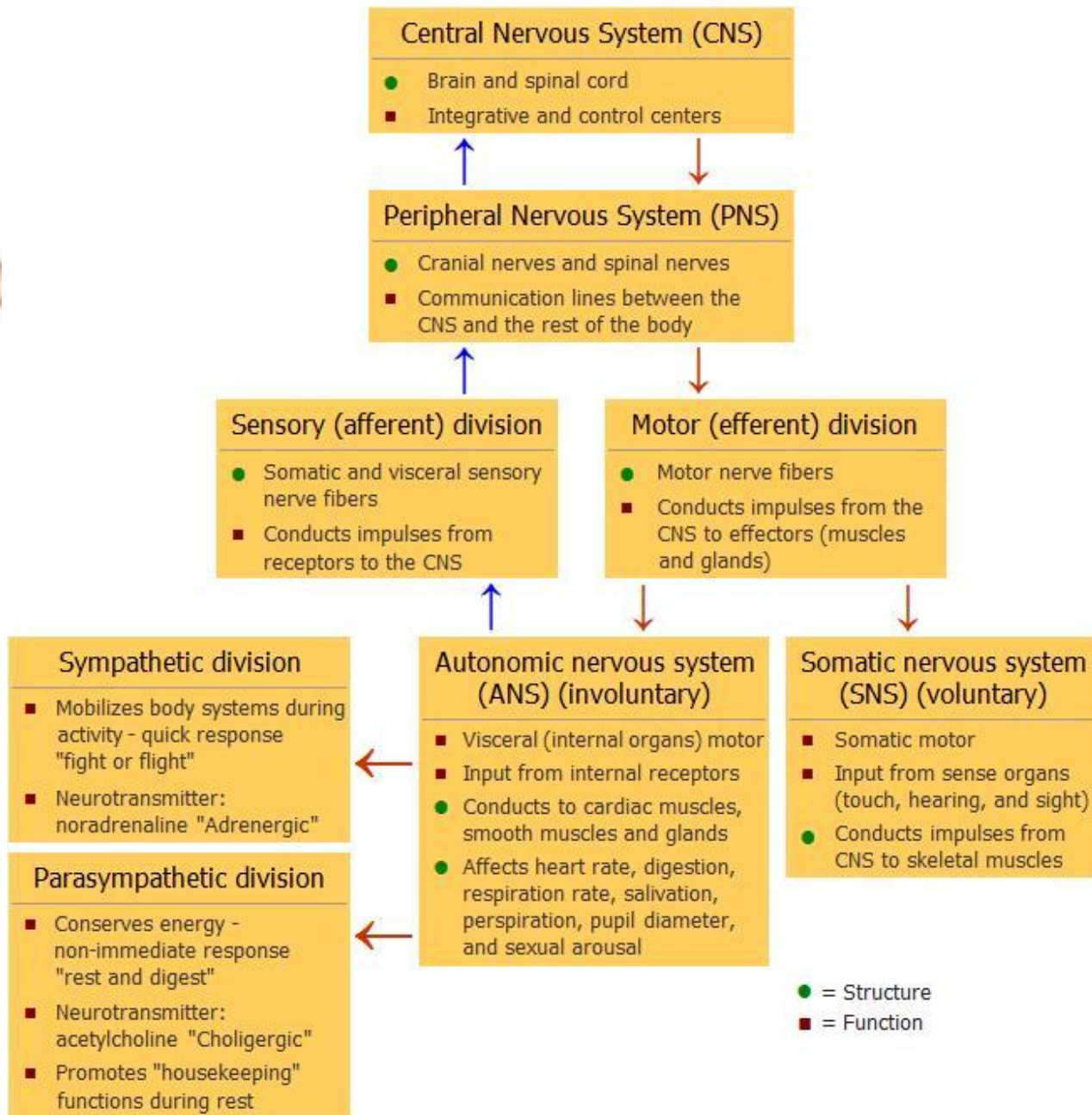
# PNS

Peripheral  
Nervous System



Branch of from CNS  
to rest of your body





# Nervous system structure

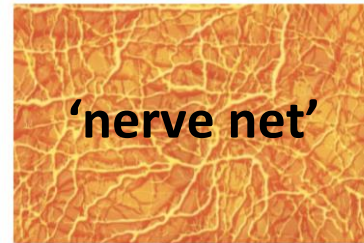
## The Nobel Prize in Physiology or Medicine 1906



Camillo Golgi  
Prize share: 1/2

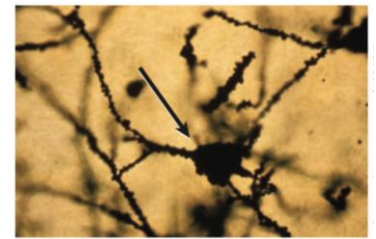


Santiago Ramón y Cajal  
Prize share: 1/2



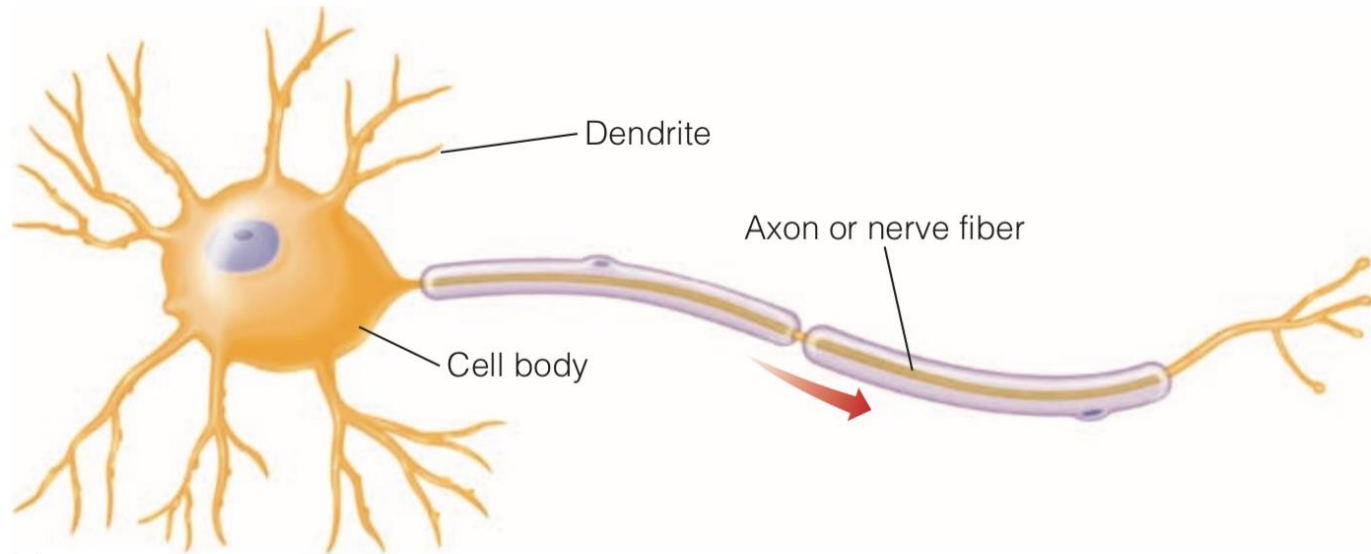
'nerve net'

reticular theory vs neuron doctrine

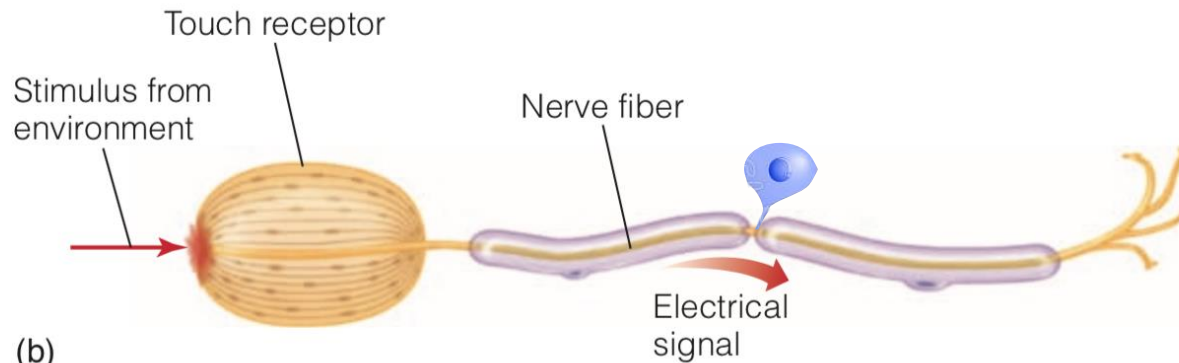




# Nervous system structure

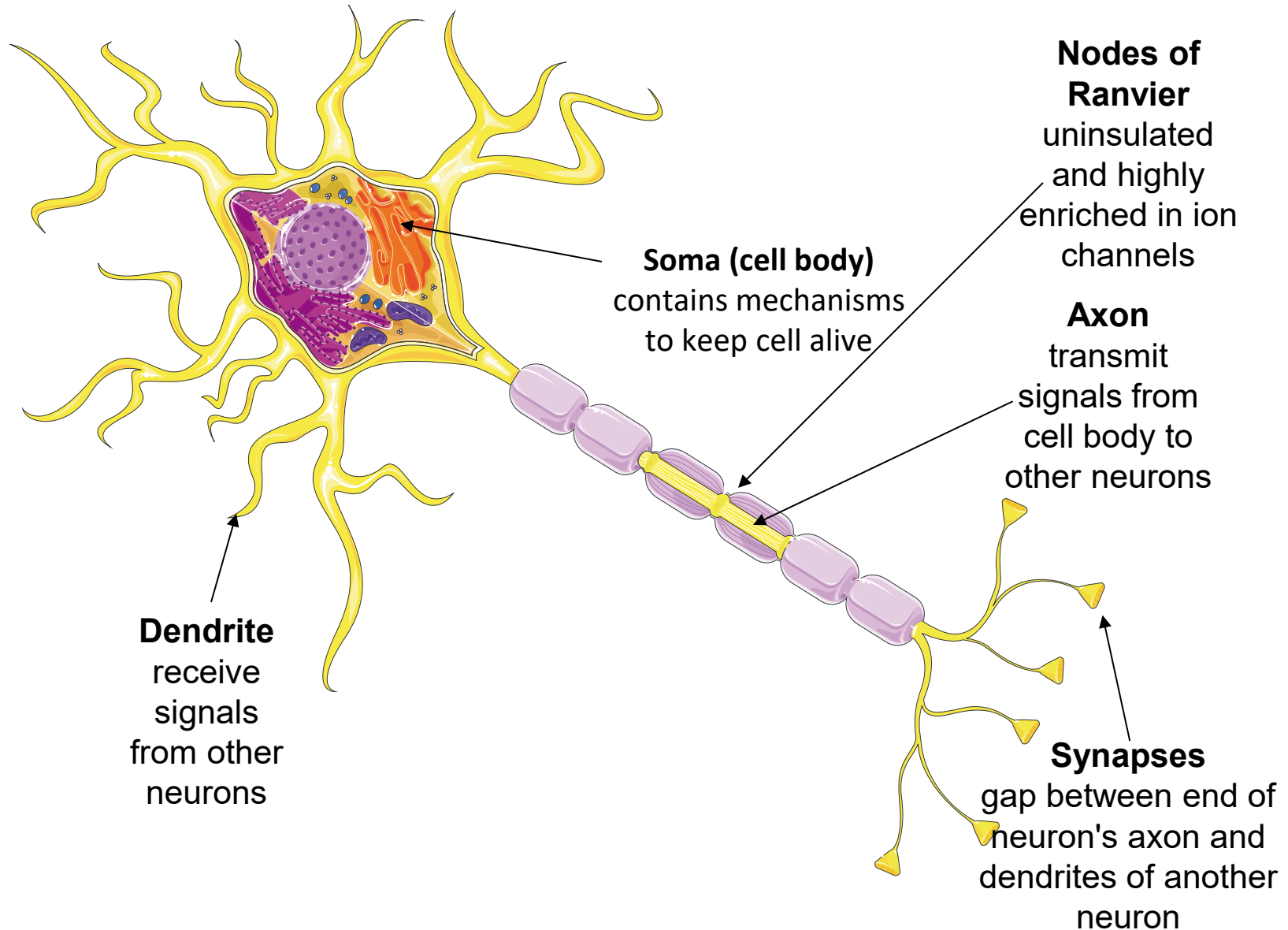


(a)

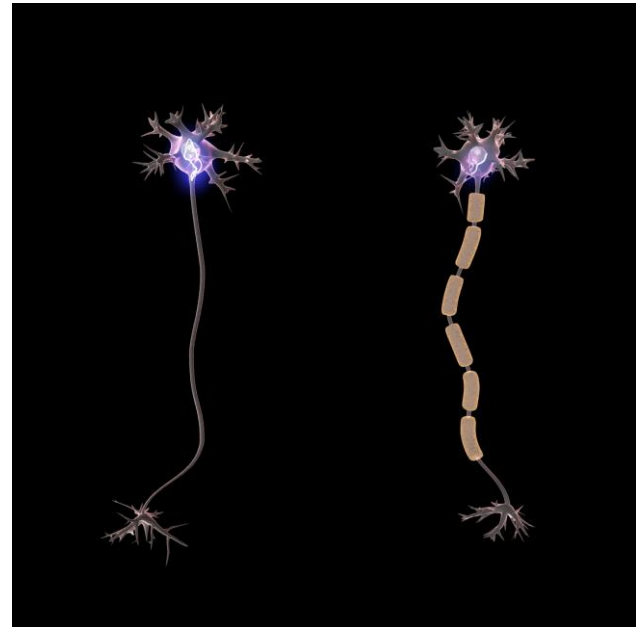
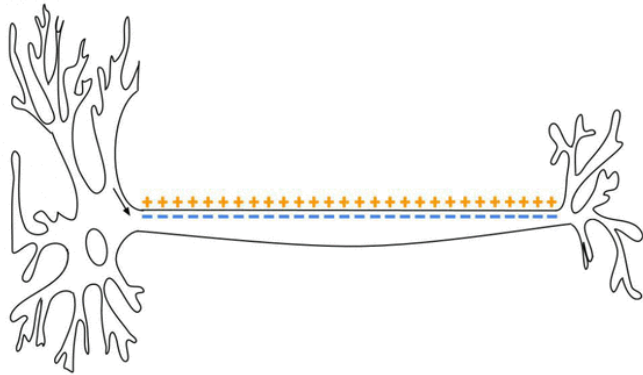


(b)

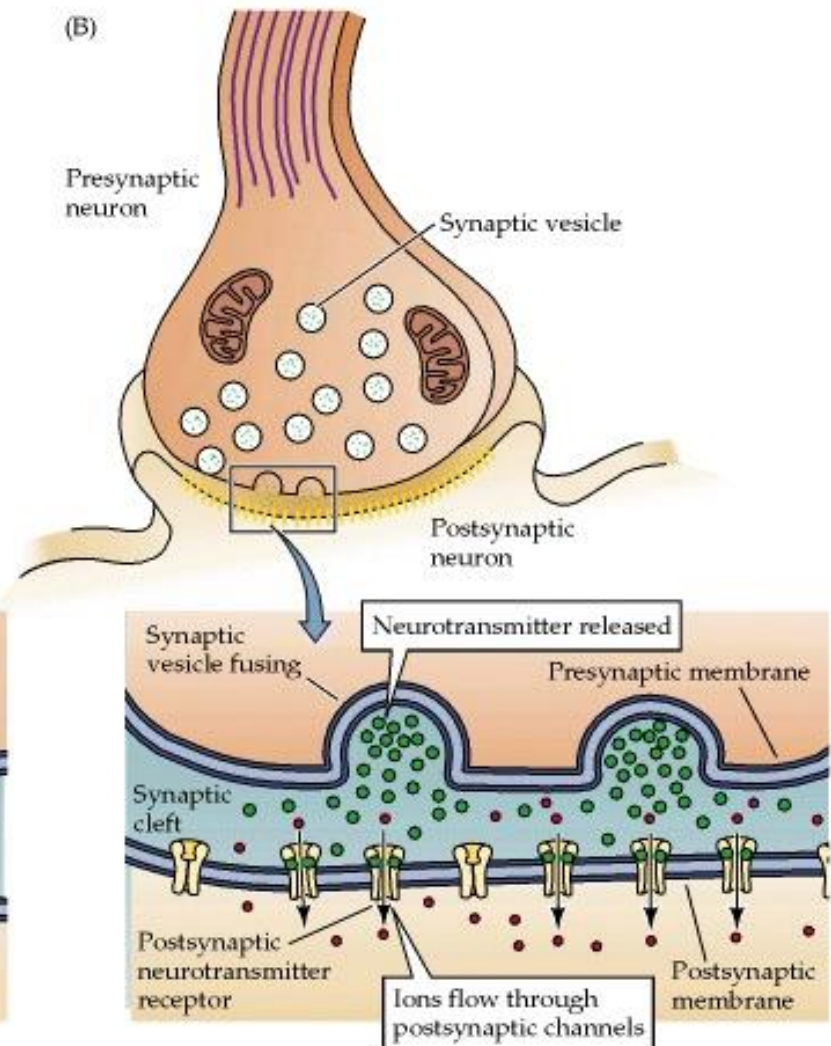
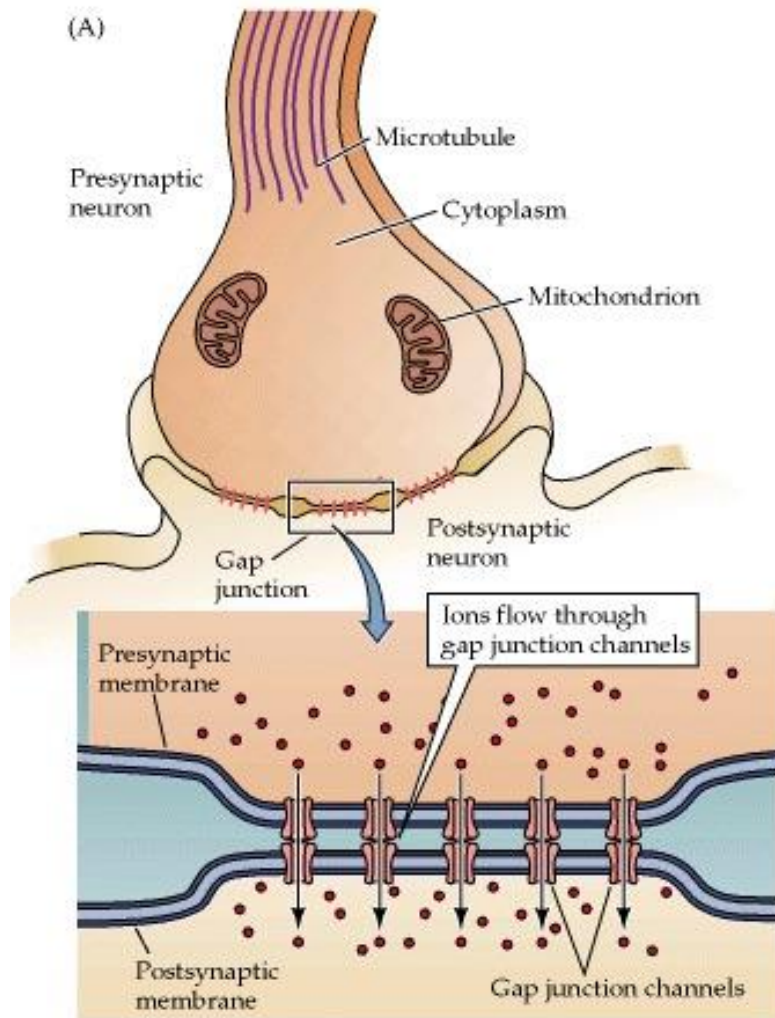
# Neuron



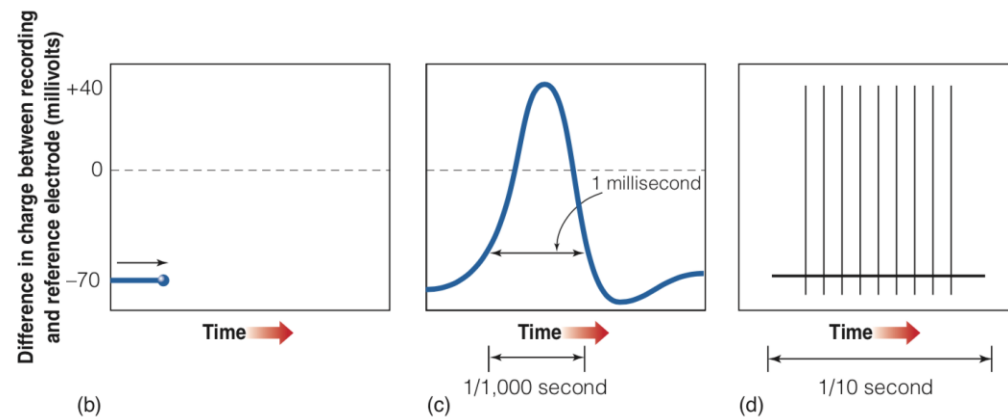
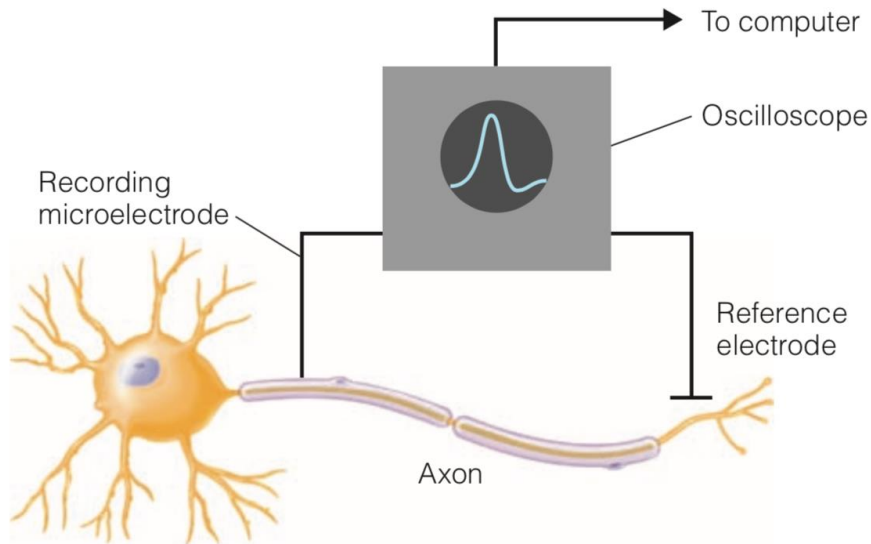
# Neuron



# Synapse



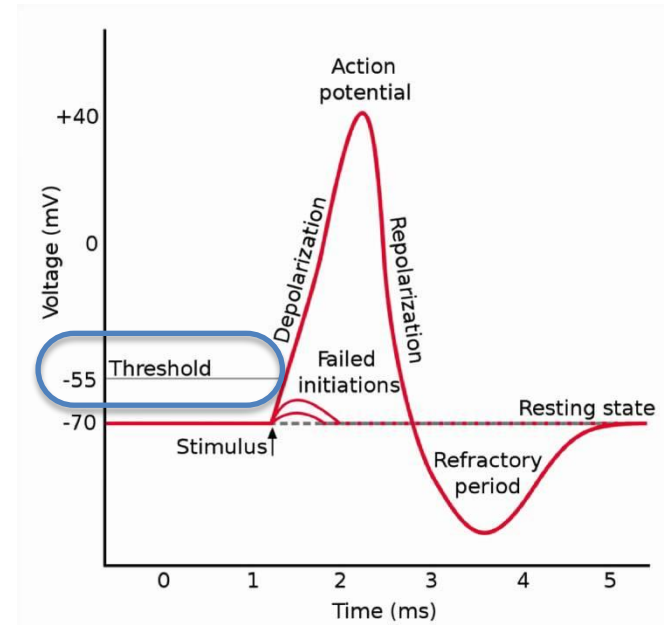
# Signals in Neurons



NOBEL PRIZE JOINT AWARD.—Sir Charles Sherrington (left) and Professor Edgar Douglas Adrian, jointly awarded the Nobel Prize for 1932 in Medicine and Physiology for their discoveries in connection with the functions of the neurons.

# Action Potentials

- nerve impulses
- resting-membrane potential - 70 mV
- a temporary shift (from negative to positive) in the neuron's membrane potential caused by ions suddenly flowing in and out of the neuron

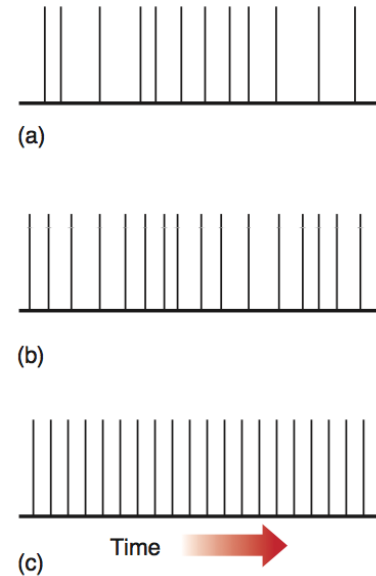


Nobel Prize in Physiology or Medicine 1963 was awarded jointly to Sir John Eccles, Alan Hodgkin and Andrew Huxley "for their discoveries concerning the ionic mechanisms involved in excitation and inhibition ... in the nerve cell membrane."



# Action Potentials

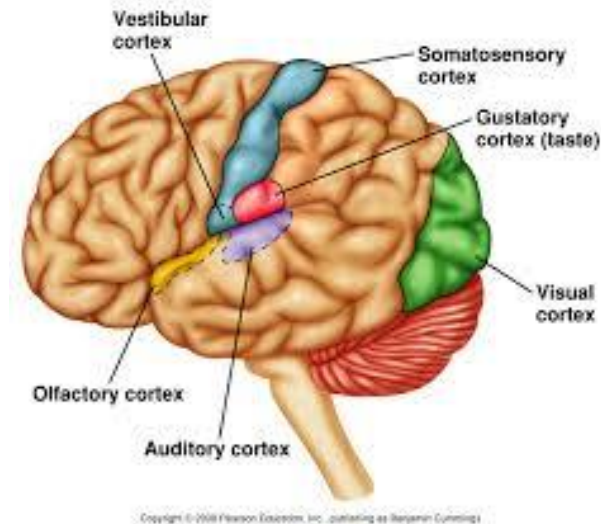
- Input is analog (membrane potential is continuous valued) and Output is discrete (neuron spikes or not)
- ex: how does firing of a neuron from a receptor in the skin change with increase in pressure applied?



stimulus intensity  
=  
rate of neural firing

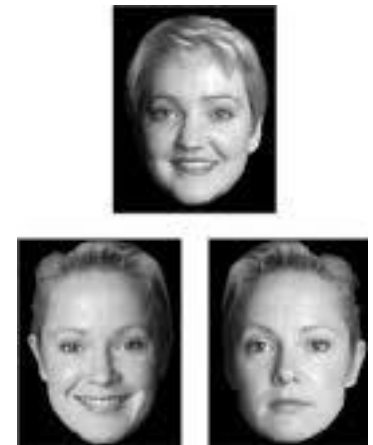
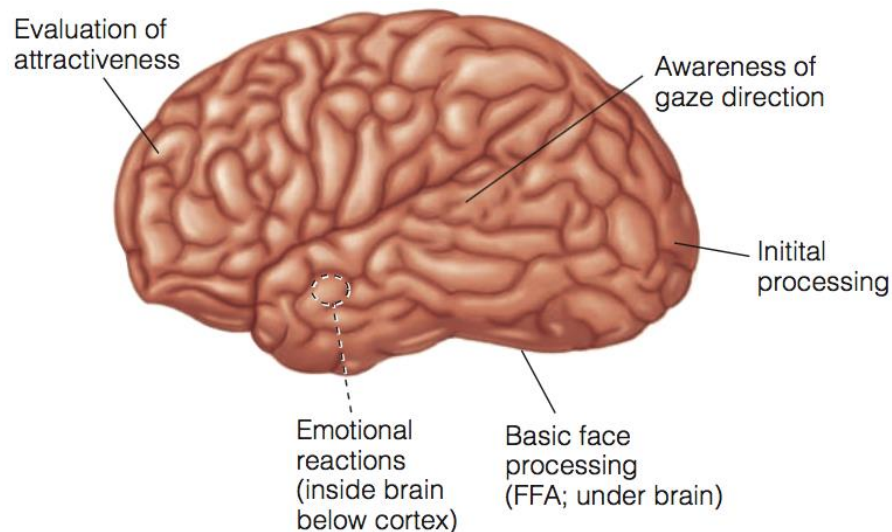
# Action Potentials to Perception

- if output is always binary then how does the brain distinguish stimuli?
- Localization of function/modularity
  - primary receiving areas
    - ex: temporal (auditory), occipital (vision), parietal (touch),etc..
  - specific sensory functional areas
    - ex: Wernicke's area, fusiform face area (in addition to other areas)



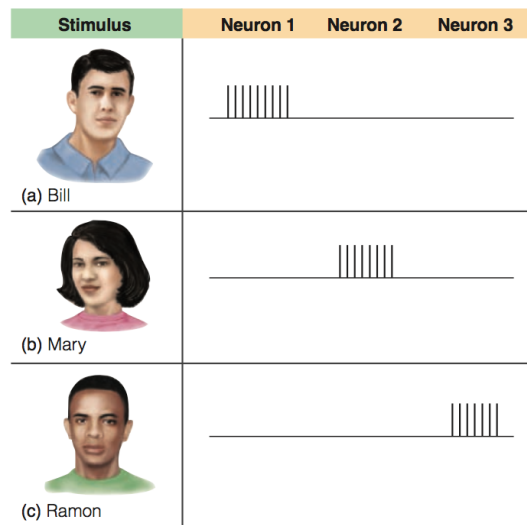
# Localization vs distributed processing

- specific functions are processed by many different areas that are associated with *representation*
  - ex: face perception - *distributed processing*

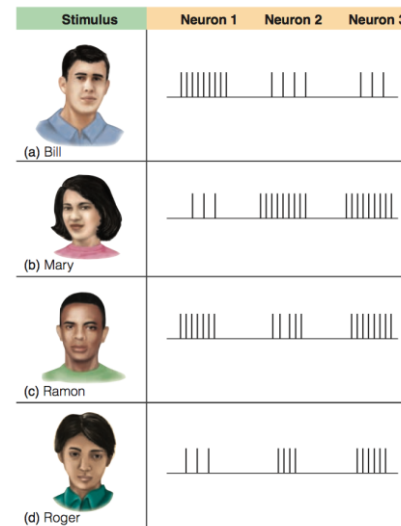


# Neural Code

- *representation/neural code/schema*
  - specificity coding vs distributed coding



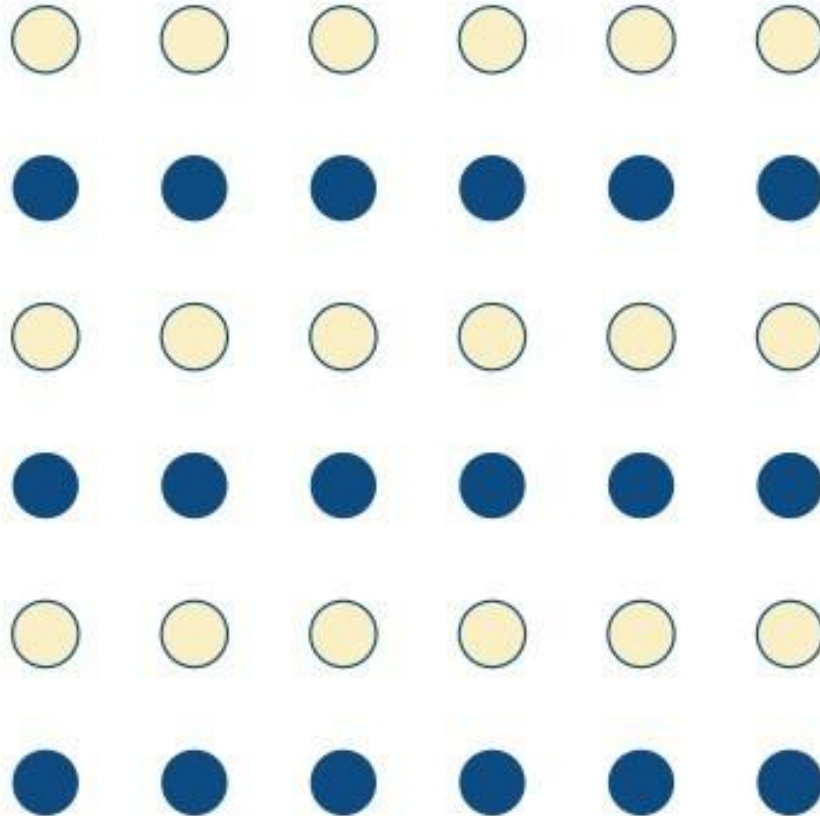
● **FIGURE 2.21** How faces could be coded by specificity coding. Each face causes one specialized neuron to respond. (Source: B. Goldstein, *Sensation and Perception* 8th ed., Fig. 2.21, p. 36. Copyright © 2010 Wadsworth, a part of Cengage Learning. Reproduced with permission. [www.cengage.com/permissions](http://www.cengage.com/permissions).)



● **FIGURE 2.22** How faces could be coded by distributed coding. Each face causes all the neurons to fire, but the pattern of firing is different for each face. One advantage of this method of coding is that many faces could be represented by the firing of the three neurons. (Source: B. Goldstein, *Sensation and Perception*, 8th ed., Fig. 2.23, p. 38. Copyright © 2010 Wadsworth, a part of Cengage Learning. Reproduced with permission. [www.cengage.com/permissions](http://www.cengage.com/permissions).)

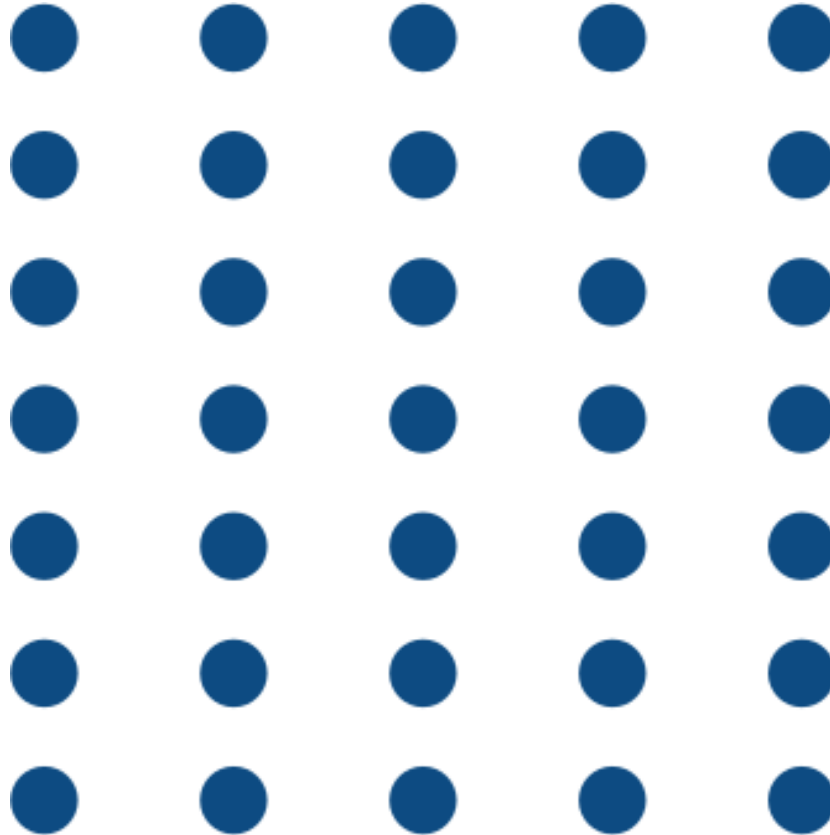
# Gestalt Principles of Organization of Visual Perception

- Similarity



# Organizational Rules of Visual Perception

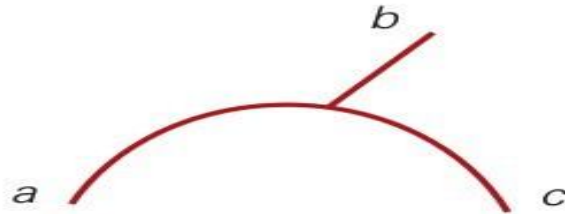
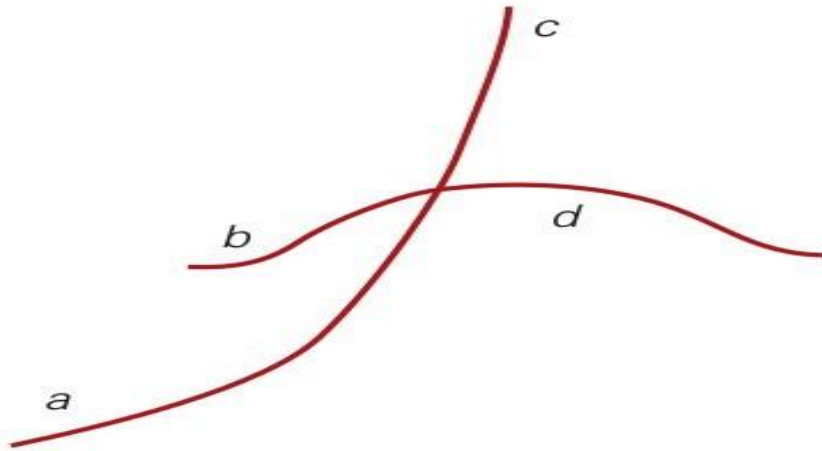
- Proximity





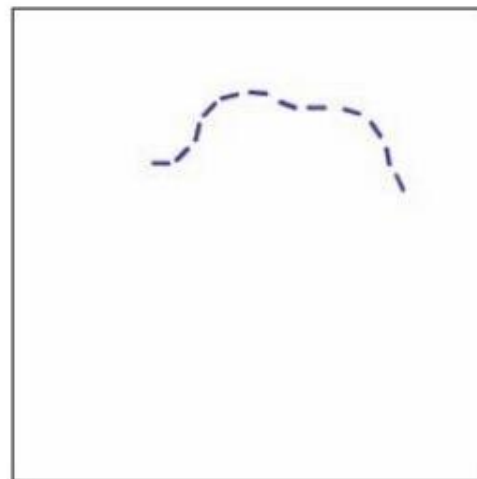
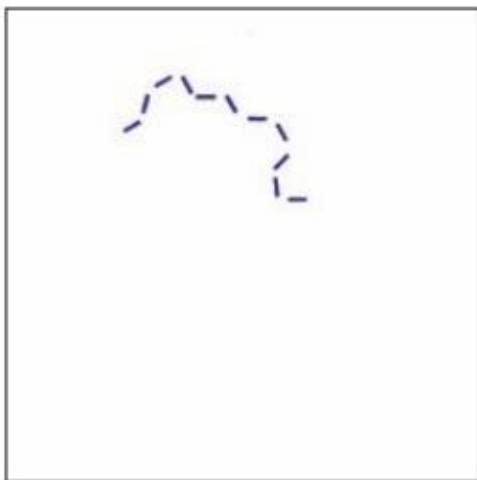
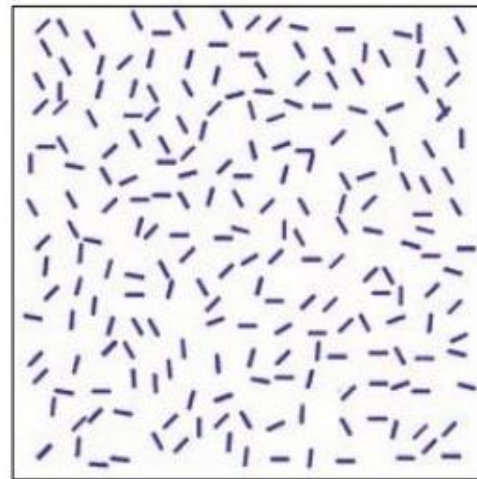
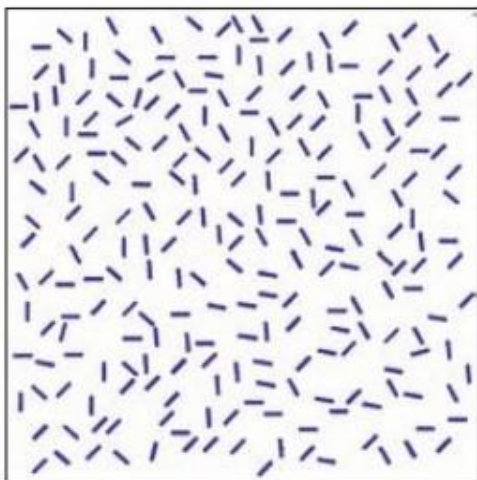
# Organizational Rules of Visual Perception

- Continuation



# Organizational Rules of Visual Perception

- Contour Saliency



Object Recognition depends on the separation of foreground and background in a scene



Expectation and Perceptual task play a critical role in what is seen

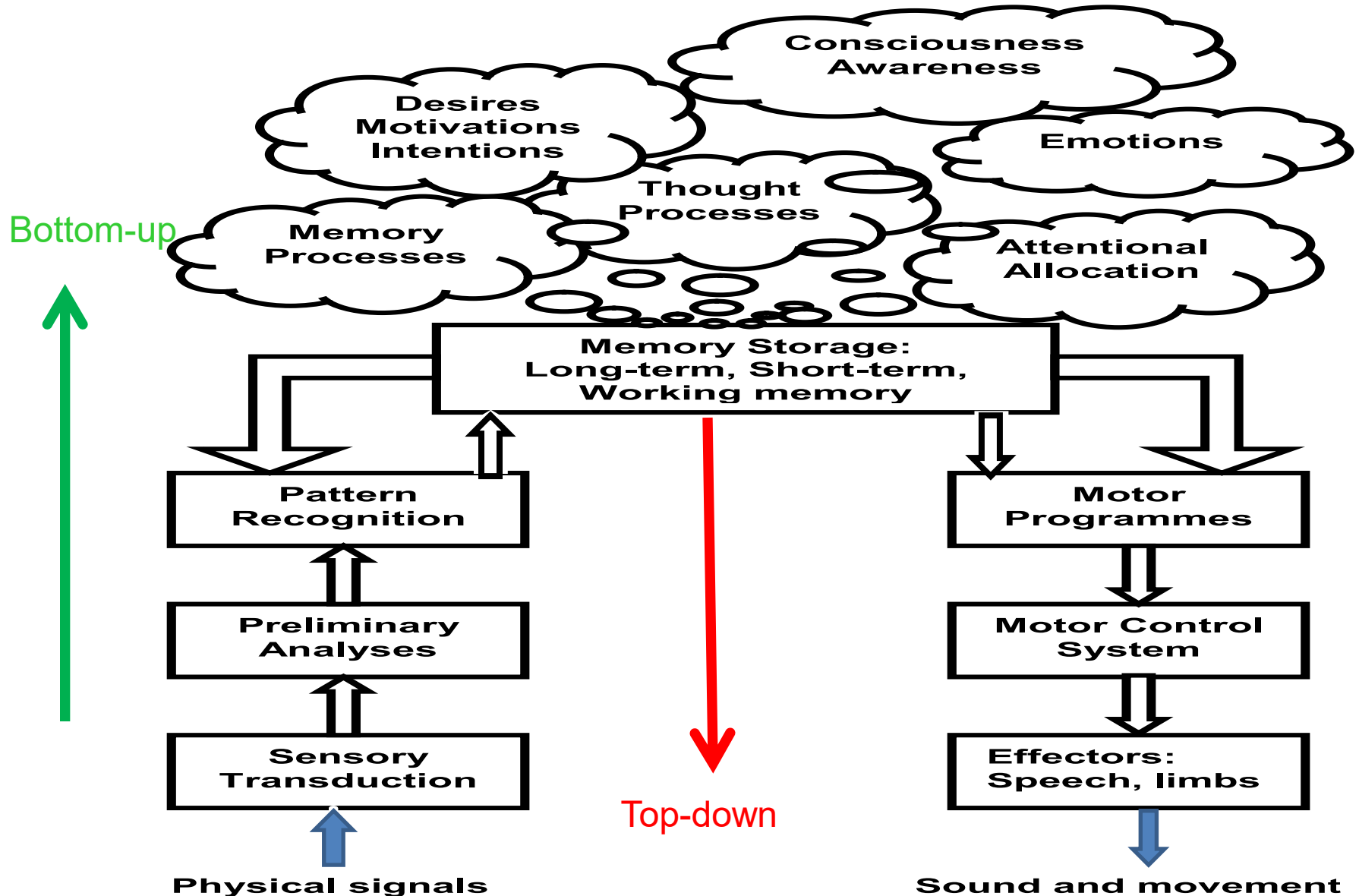




Expectation and Perceptual task play a critical role in what is seen



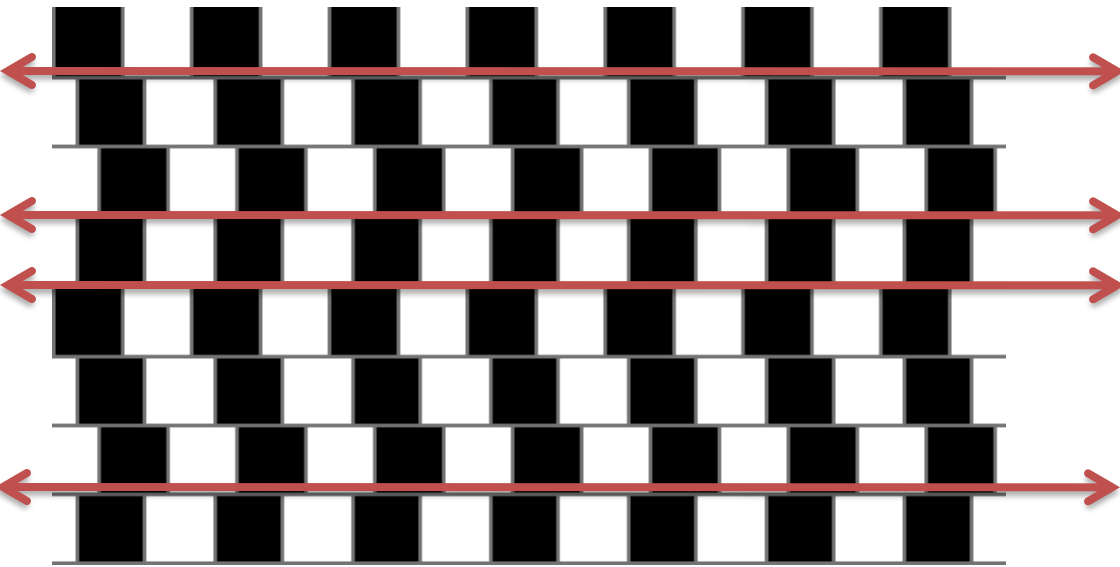
# *Human Information Processing System*



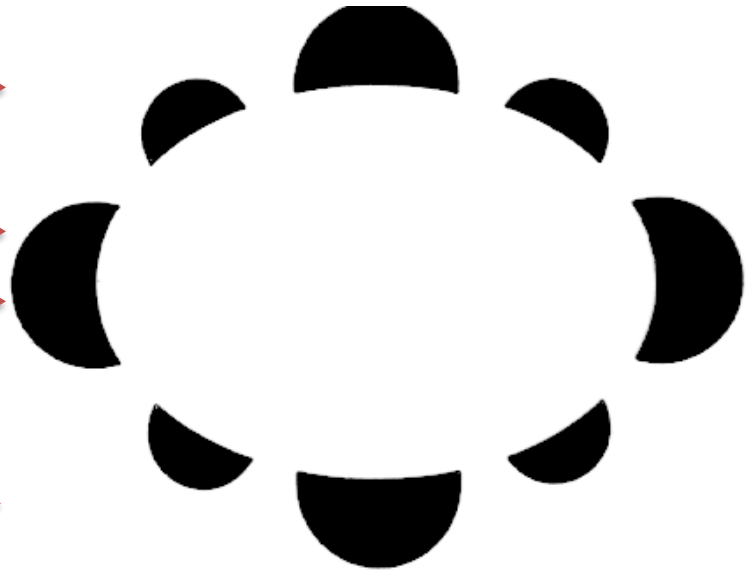


# *Bottom-up or Top-down Processing?*

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Bottom-Up ✓

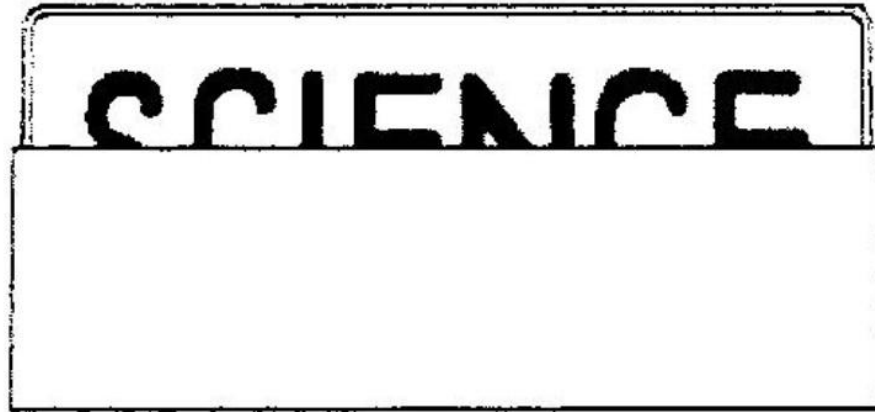


Bottom-Up ✓

# *Bottom-up or Top-down Processing?*

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What do you see?



Now what do you see?



Top-Down ✓